

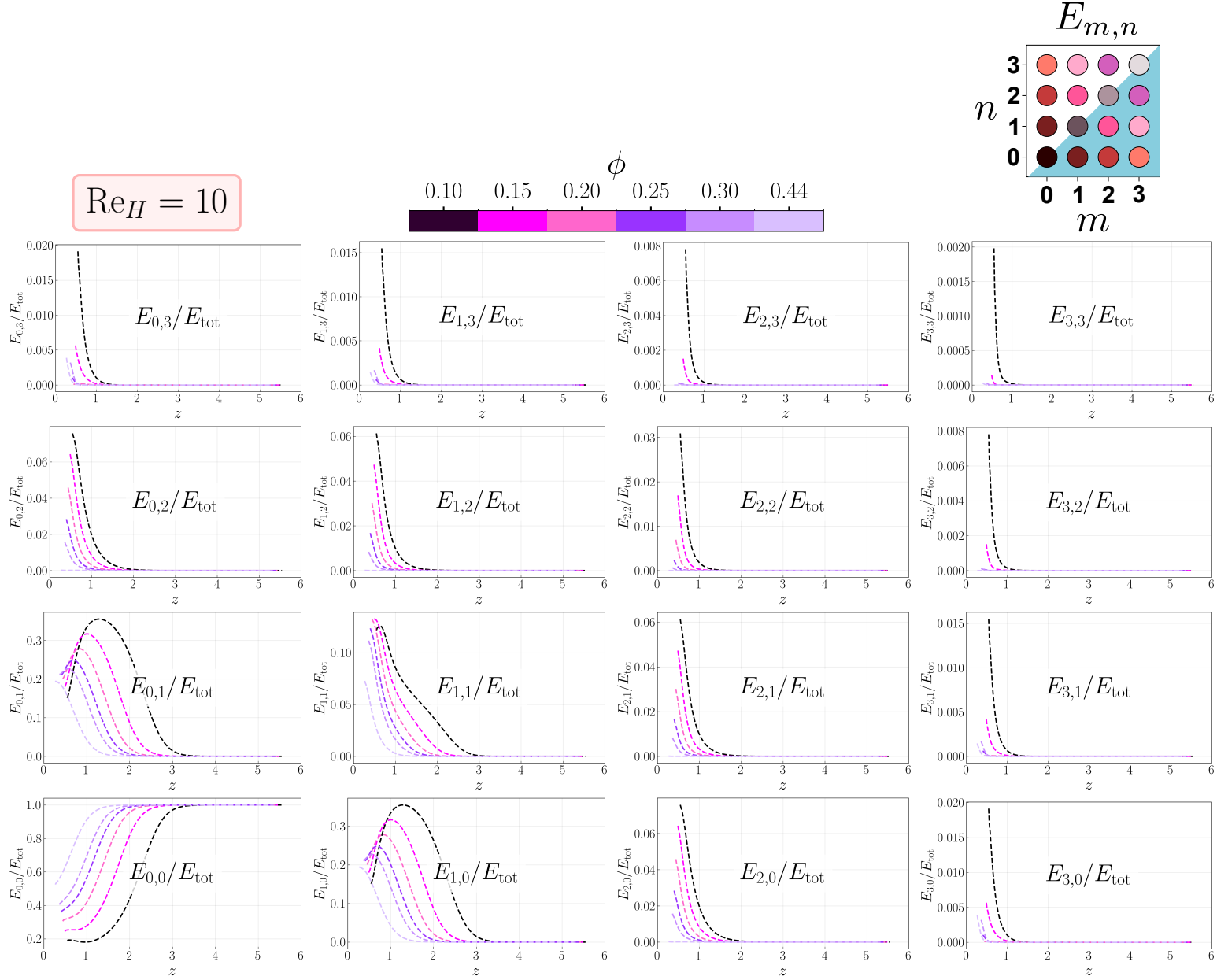


Figure B.66: Streamwise evolution of the normalized modal energy  $E_{m,n}/E_{\text{tot}}$  downstream of the SRG at  $\text{Re}_H = 10$ . Each panel shows one Fourier mode with positive indices  $m, n \in \{0, 1, 2, 3\}$ ; colors indicate the porosity  $\phi$ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property  $E_{m,n} = E_{n,m}$ .

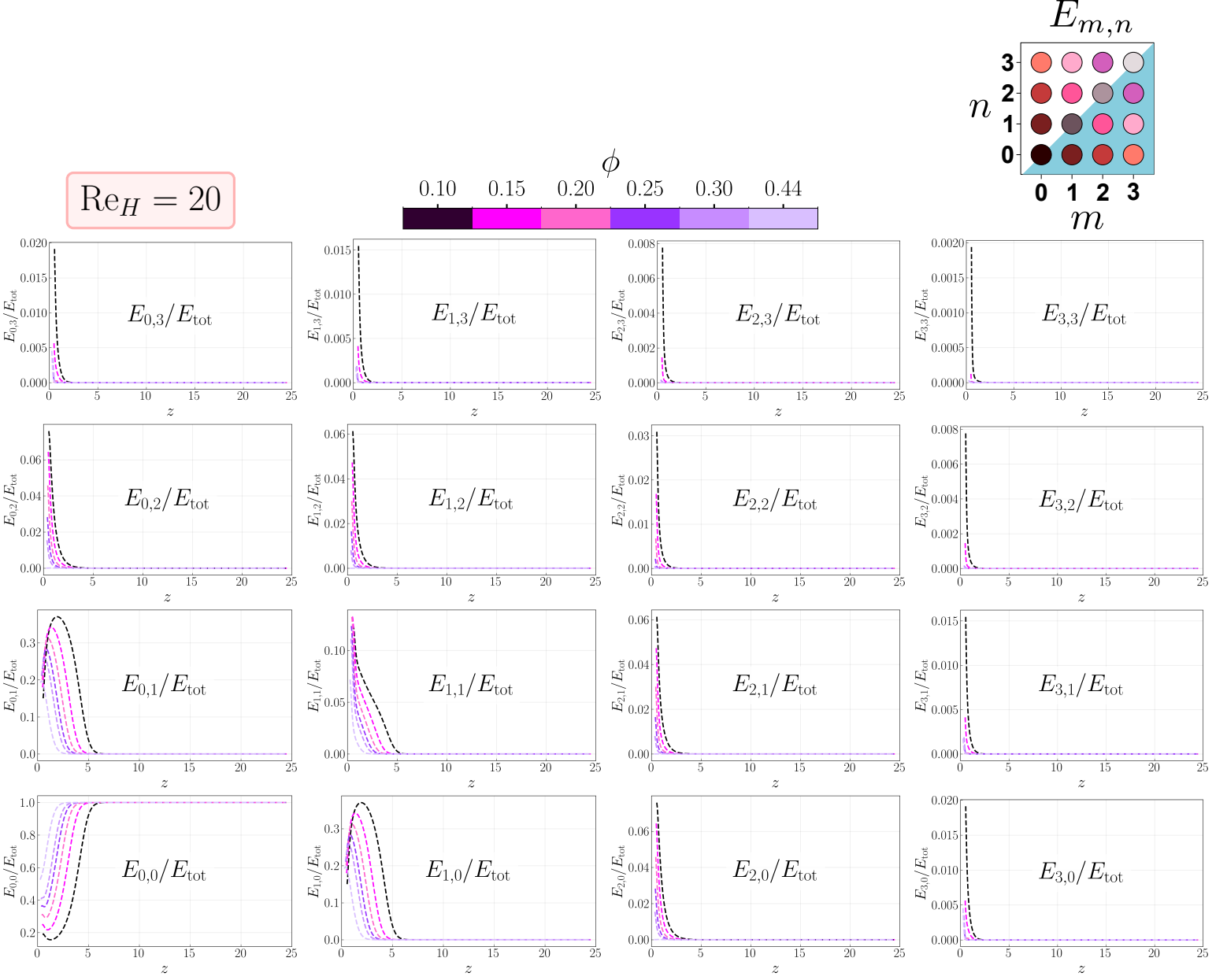


Figure B.67: Streamwise evolution of the normalized modal energy  $E_{m,n}/E_{\text{tot}}$  downstream of the SRG at  $\text{Re}_H = 20$ . Each panel shows one Fourier mode with positive indices  $m, n \in \{0, 1, 2, 3\}$ ; colors indicate the porosity  $\phi$ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property  $E_{m,n} = E_{n,m}$ .

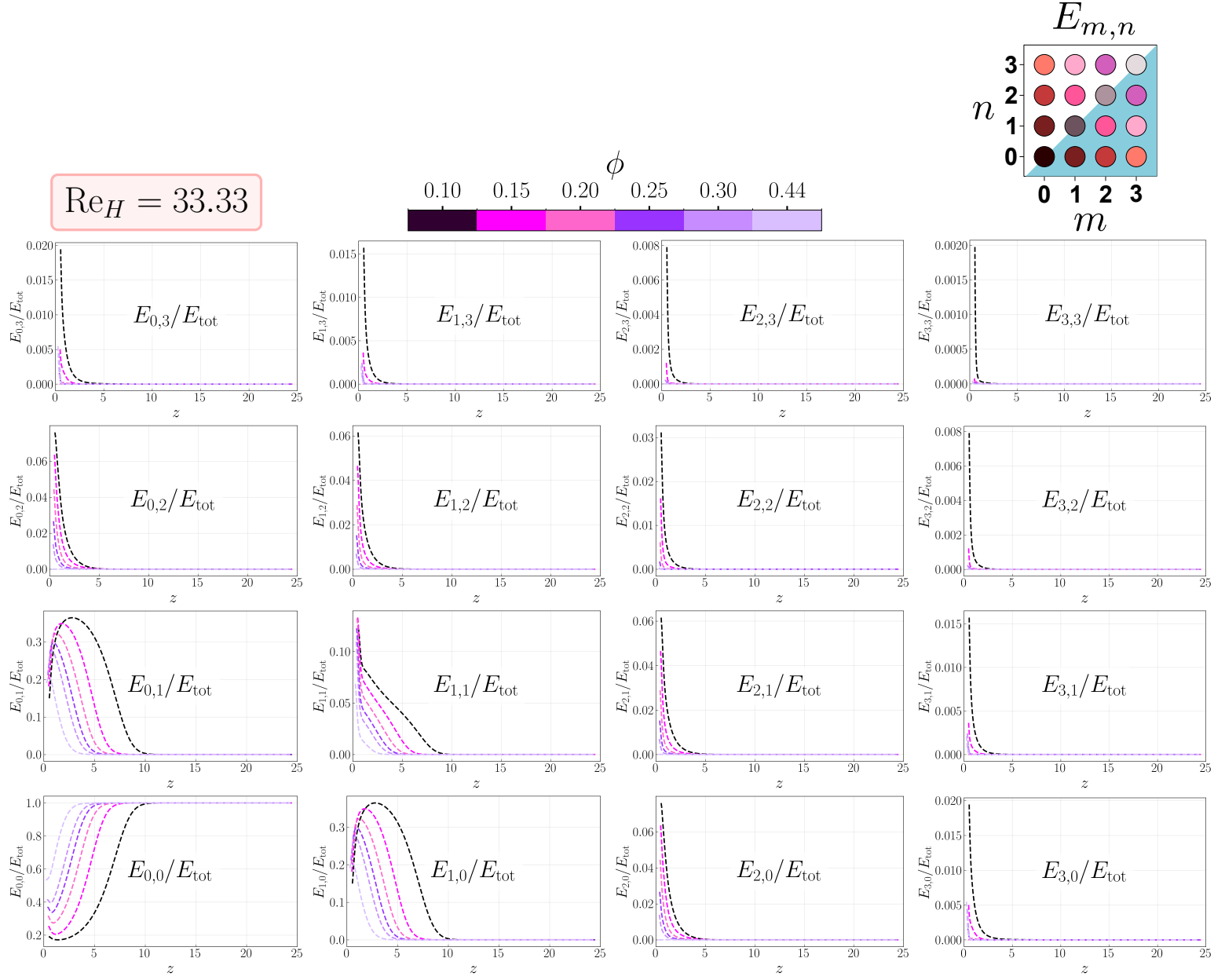


Figure B.68: Streamwise evolution of the normalized modal energy  $E_{m,n}/E_{\text{tot}}$  downstream of the SRG at  $\text{Re}_H = 33.33$ . Each panel shows one Fourier mode with positive indices  $m, n \in \{0, 1, 2, 3\}$ ; colors indicate the porosity  $\phi$ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property  $E_{m,n} = E_{n,m}$ .

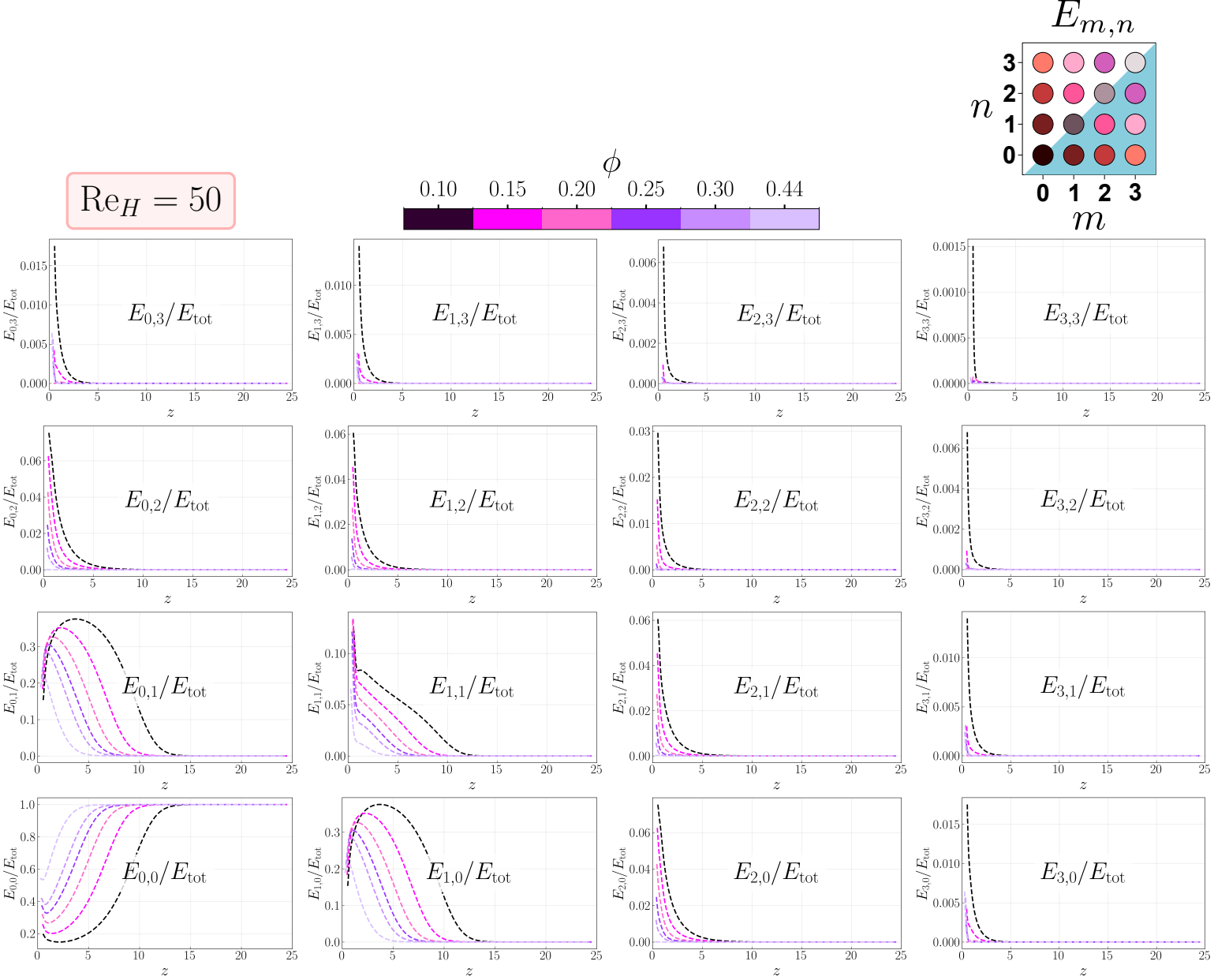


Figure B.69: Streamwise evolution of the normalized modal energy  $E_{m,n}/E_{\text{tot}}$  downstream of the SRG at  $\text{Re}_H = 50$ . Each panel shows one Fourier mode with positive indices  $m, n \in \{0, 1, 2, 3\}$ ; colors indicate the porosity  $\phi$ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property  $E_{m,n} = E_{n,m}$ .



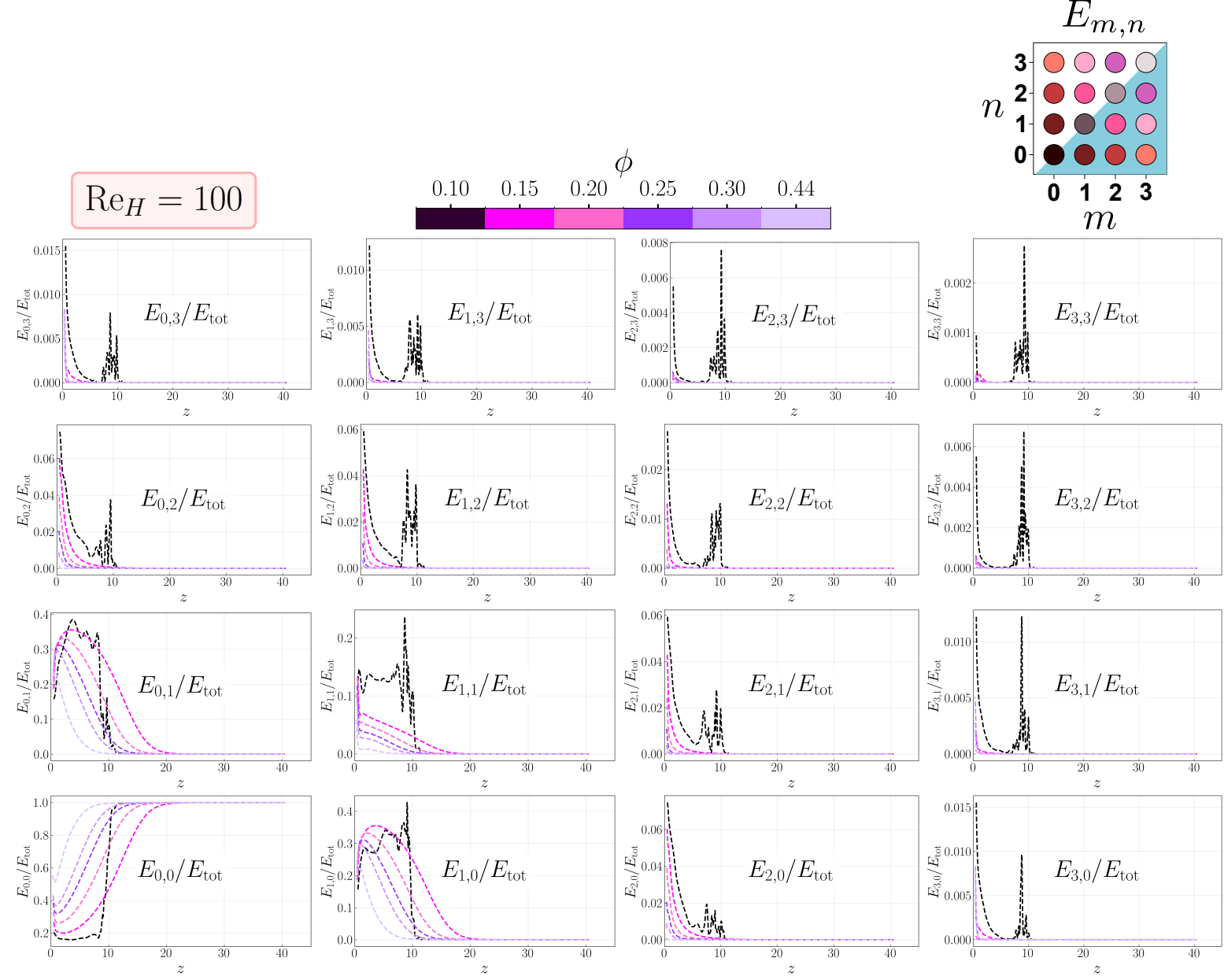


Figure B.70: Streamwise evolution of the normalized modal energy  $E_{m,n}/E_{\text{tot}}$  downstream of the SRG at  $\text{Re}_H = 100$ . Each panel shows one Fourier mode with positive indices  $m, n \in \{0, 1, 2, 3\}$ ; colors indicate the porosity  $\phi$ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes. For the steady-flow cases, the modal energy satisfies the symmetry relation  $E_{m,n} = E_{n,m}$ ; however, this symmetry is broken for the unsteady case at  $\text{Re}_H = 100$  and  $\phi = 0.10$ .

To assess the effect of modal truncation, the cross-sectional  $L_2$ -norms of the reconstructed streamwise velocity  $w$  and forcing  $F_z$  are compared with the geometry-resolved SSF fields. Here,  $\mathcal{L}_{\text{peak}}$  denotes the distance from the grid exit to the peak of the downstream corrected-velocity response.

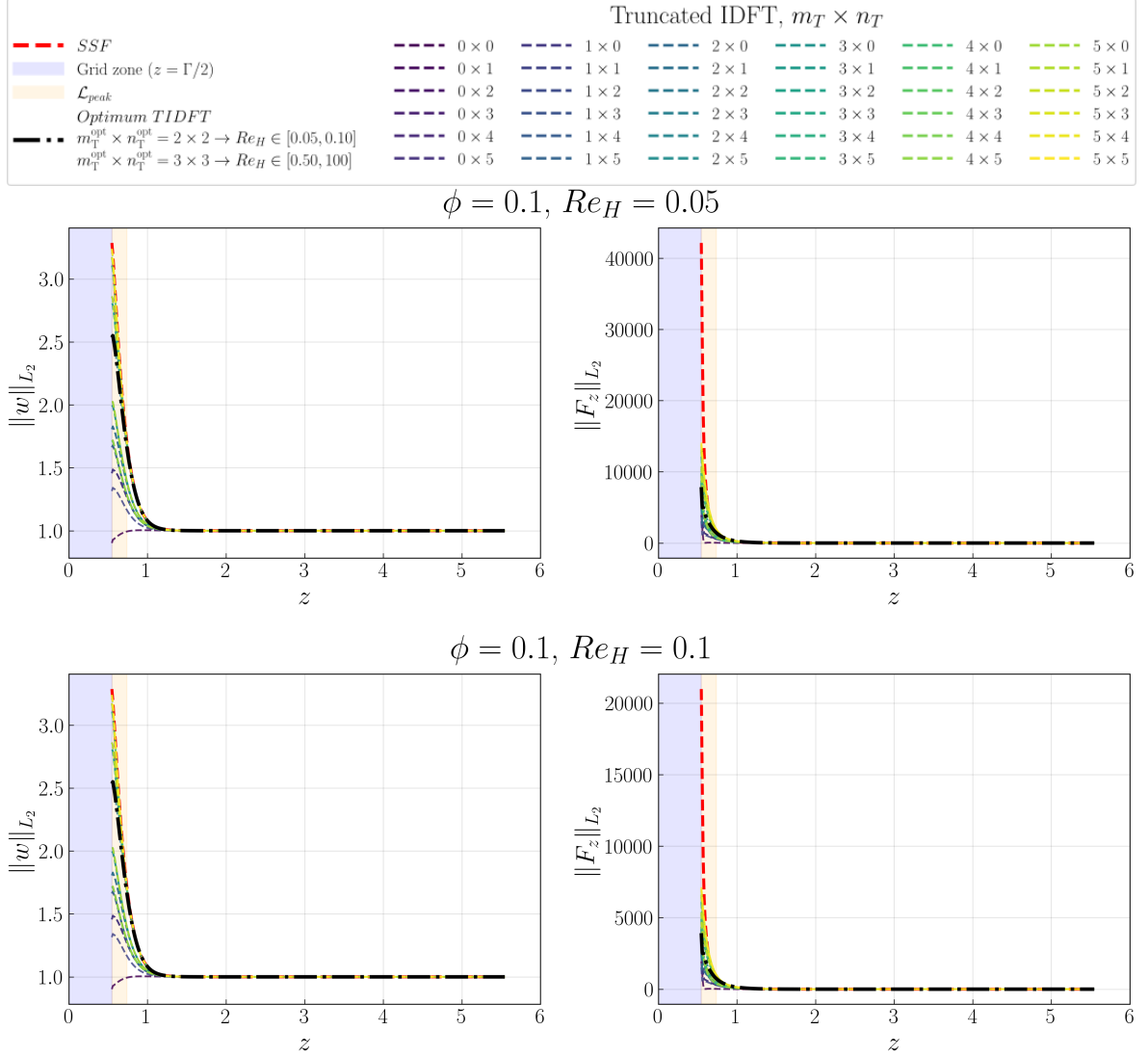


Figure B.71: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.1$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.05$ , and the lower row to  $Re_H = 0.1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

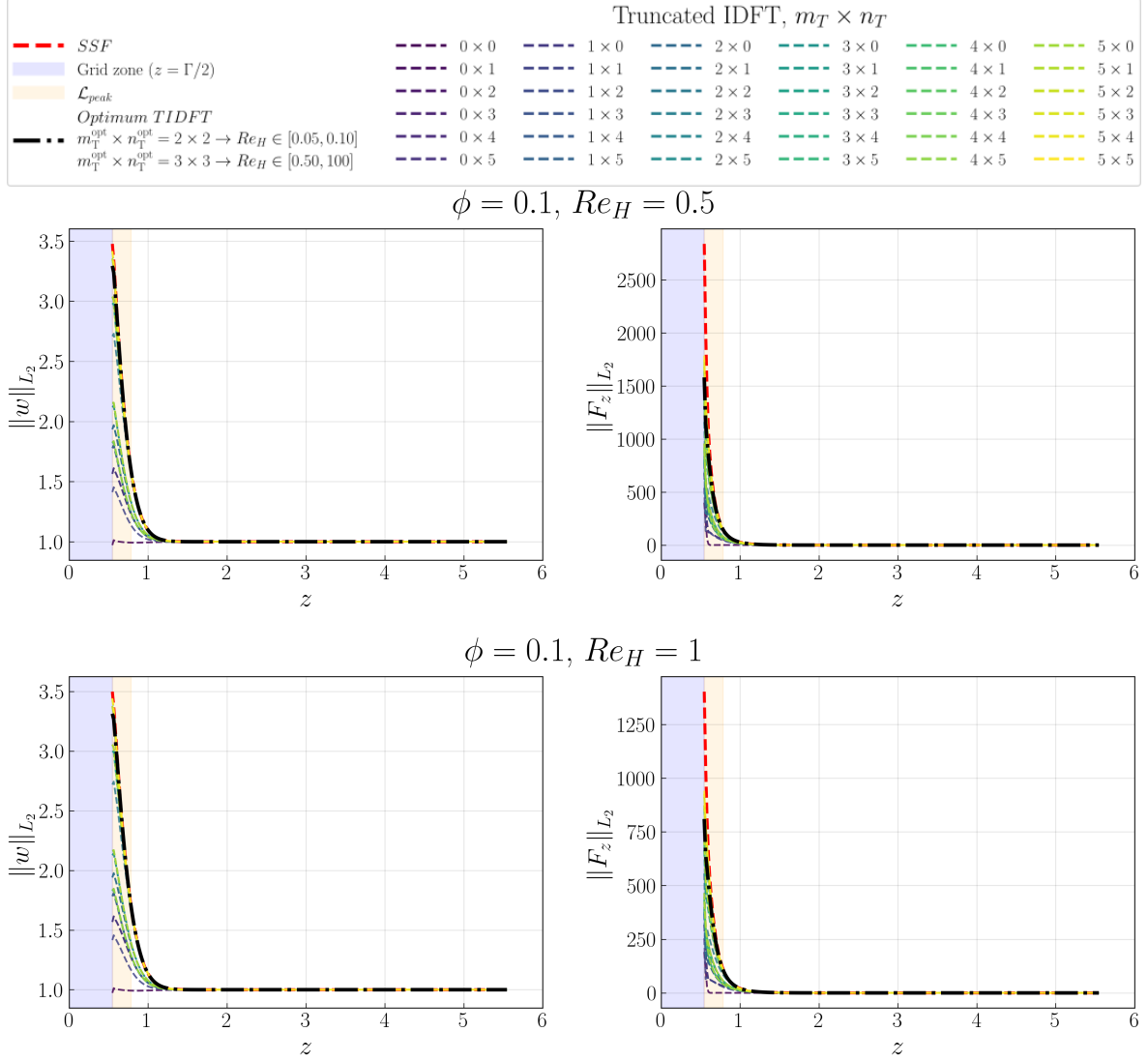


Figure B.72: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.1$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.5$ , and the lower row to  $Re_H = 1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

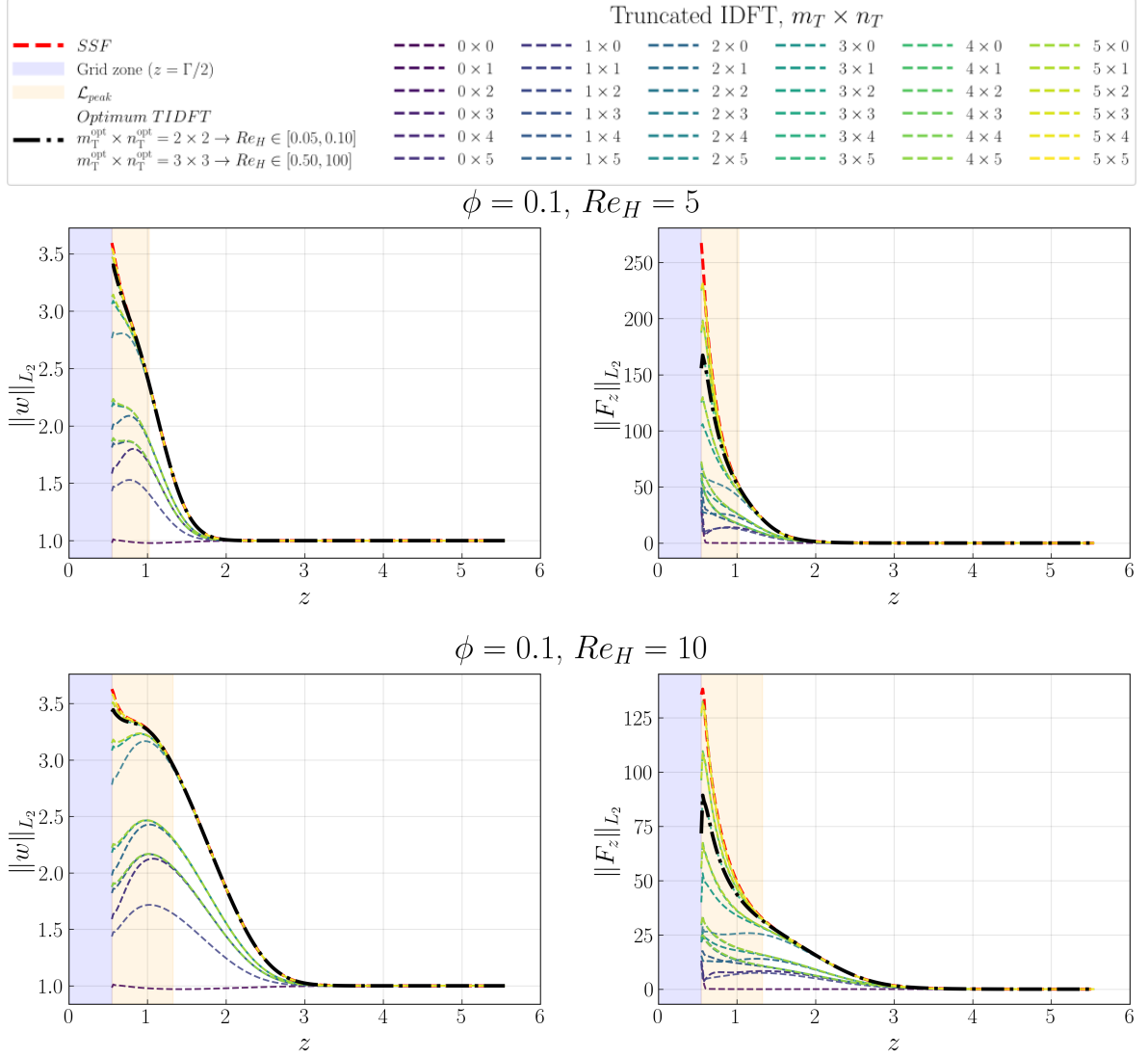


Figure B.73: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.1$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 5$ , and the lower row to  $Re_H = 10$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

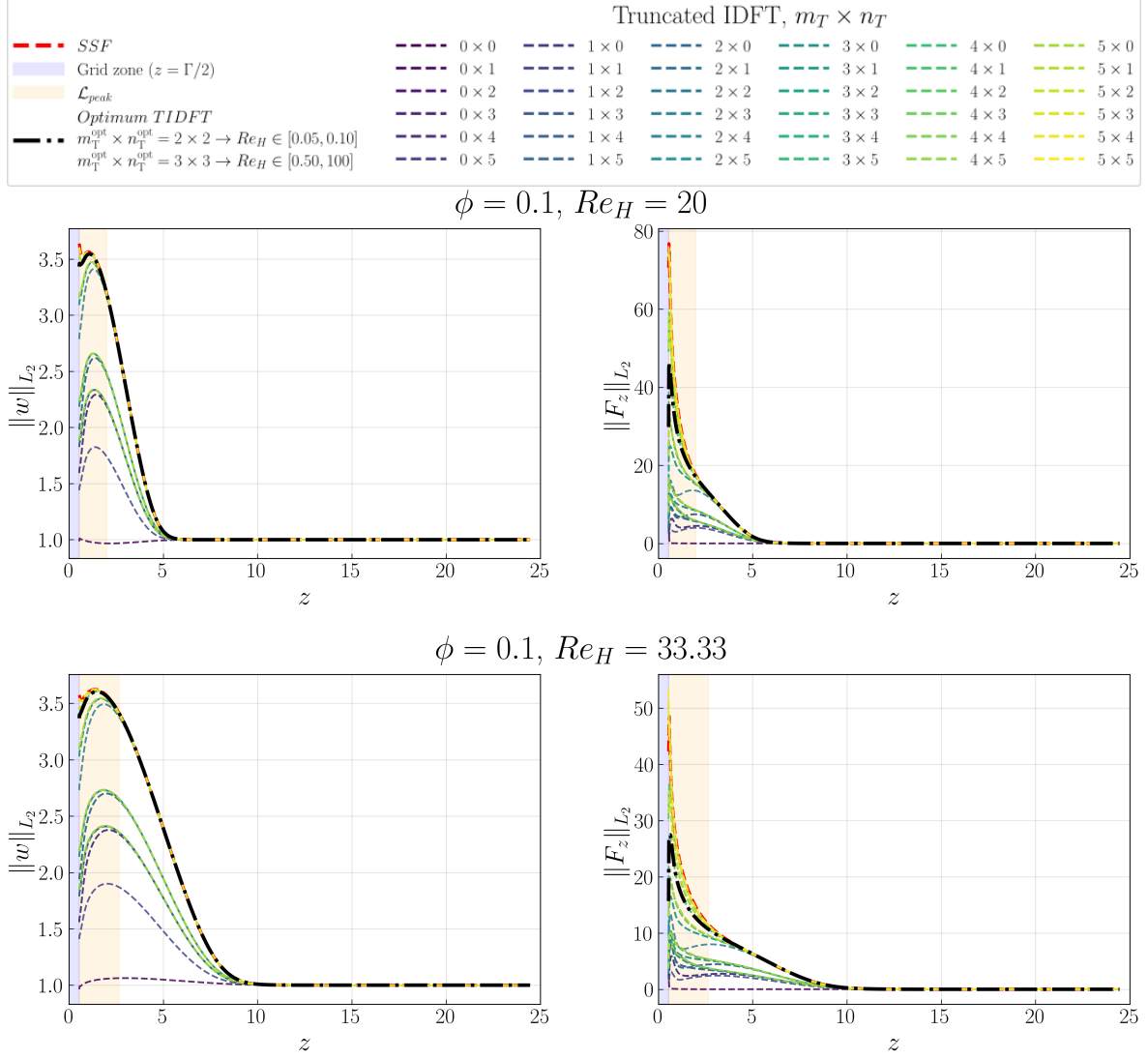


Figure B.74: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.1$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 20$ , and the lower row to  $Re_H = 33.33$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

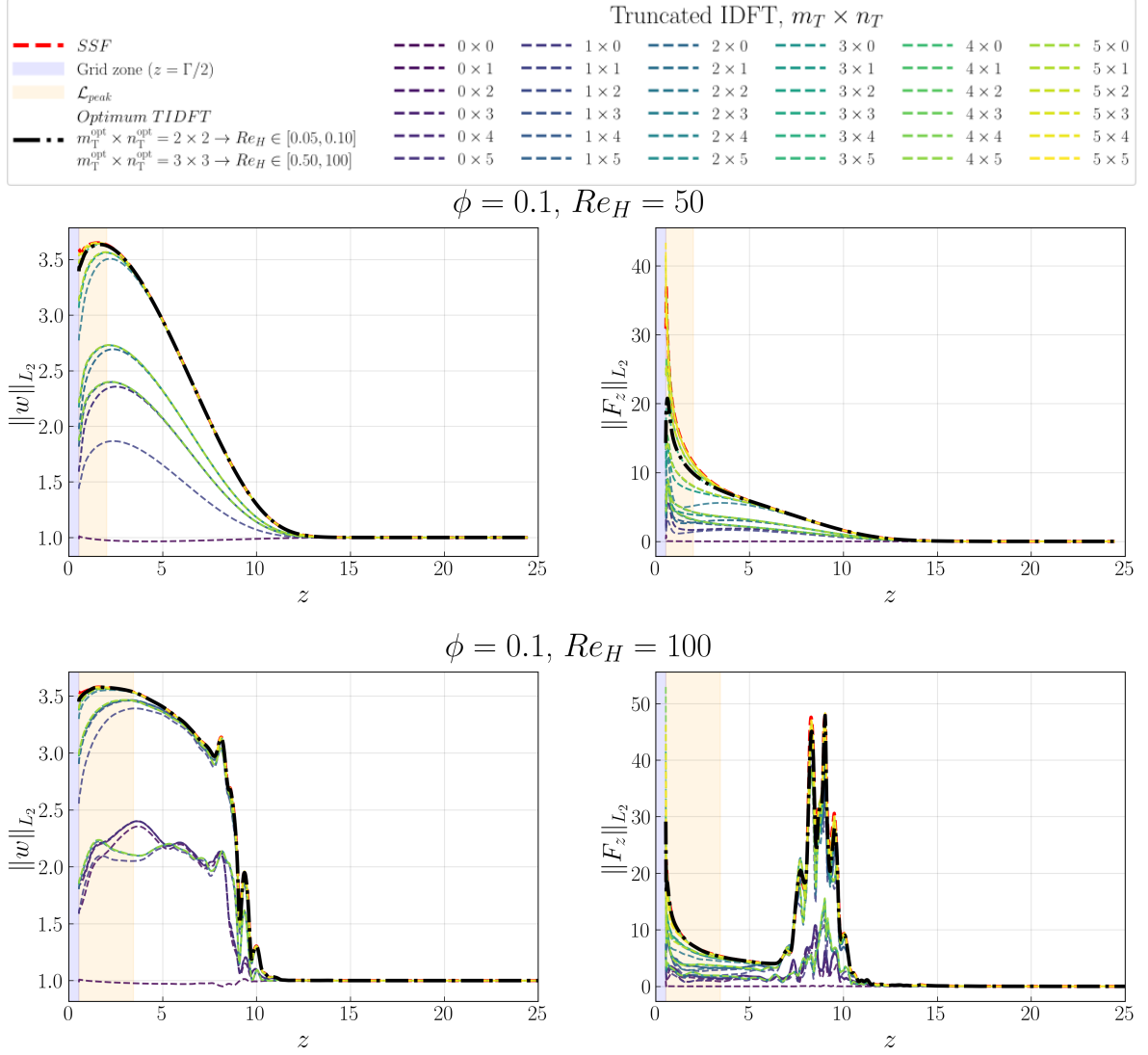


Figure B.75: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.1$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 50$ , and the lower row to  $Re_H = 100$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

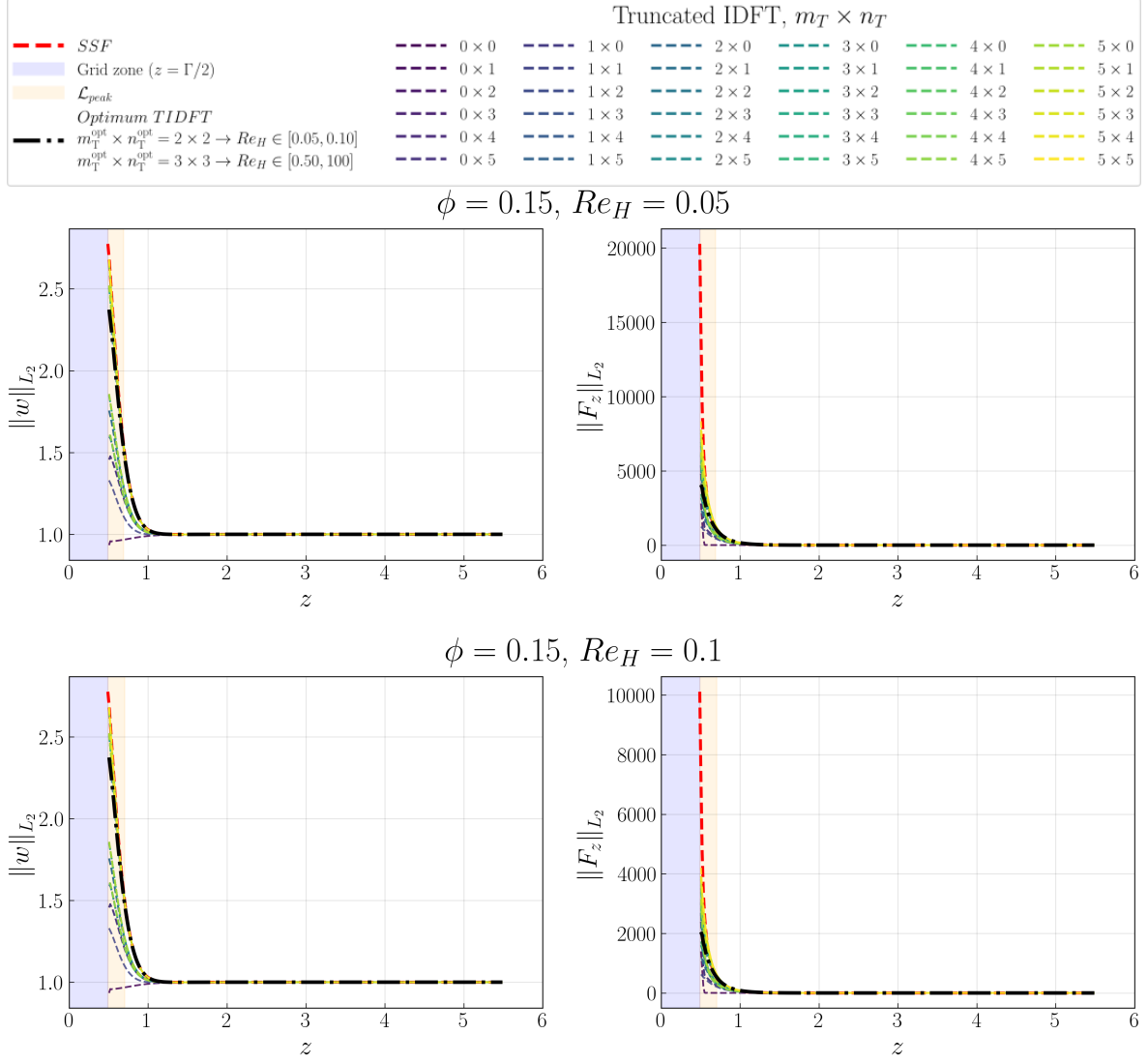


Figure B.76: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.15$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.05$ , and the lower row to  $Re_H = 0.1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .



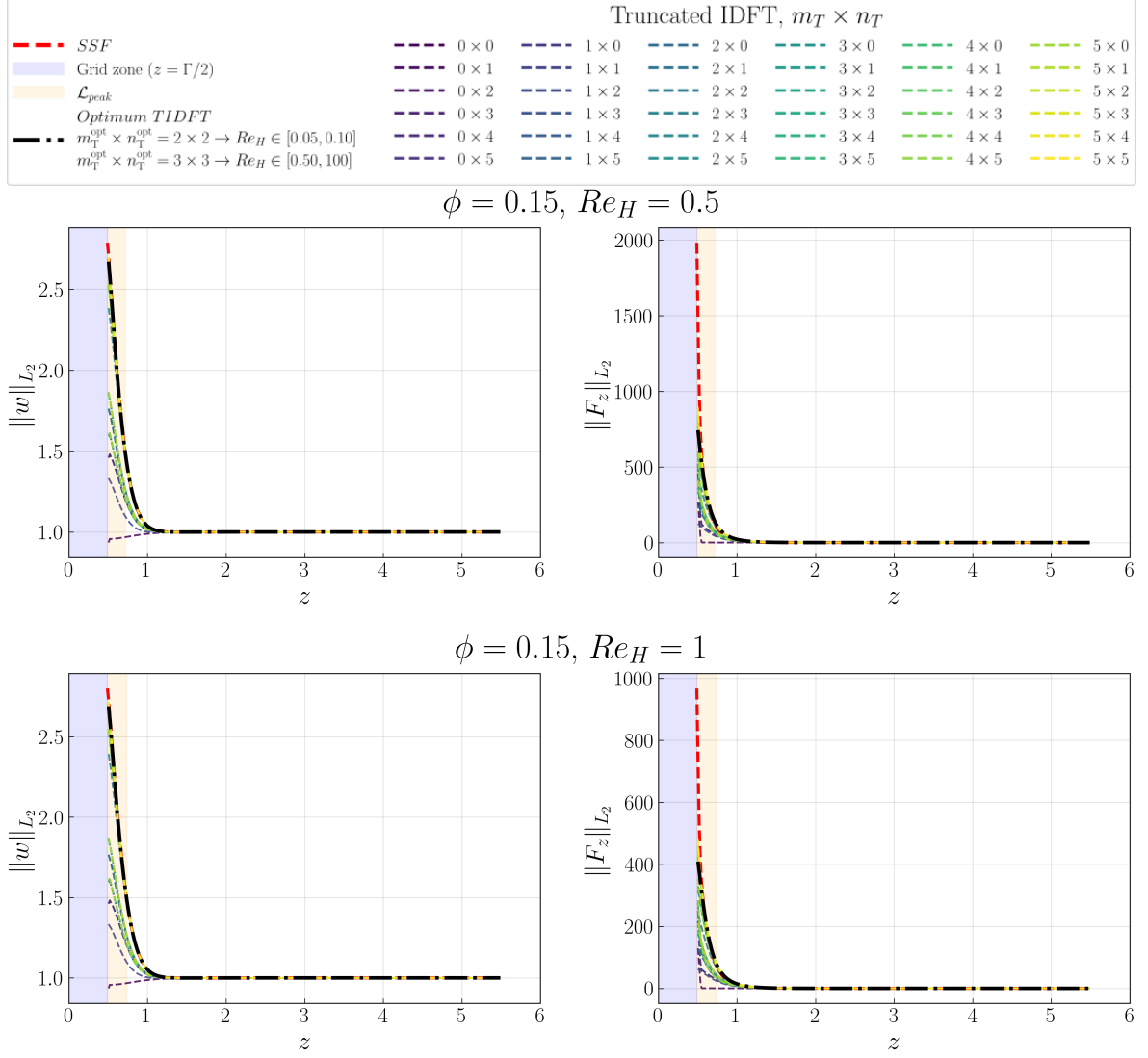


Figure B.77: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.15$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.5$ , and the lower row to  $Re_H = 1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

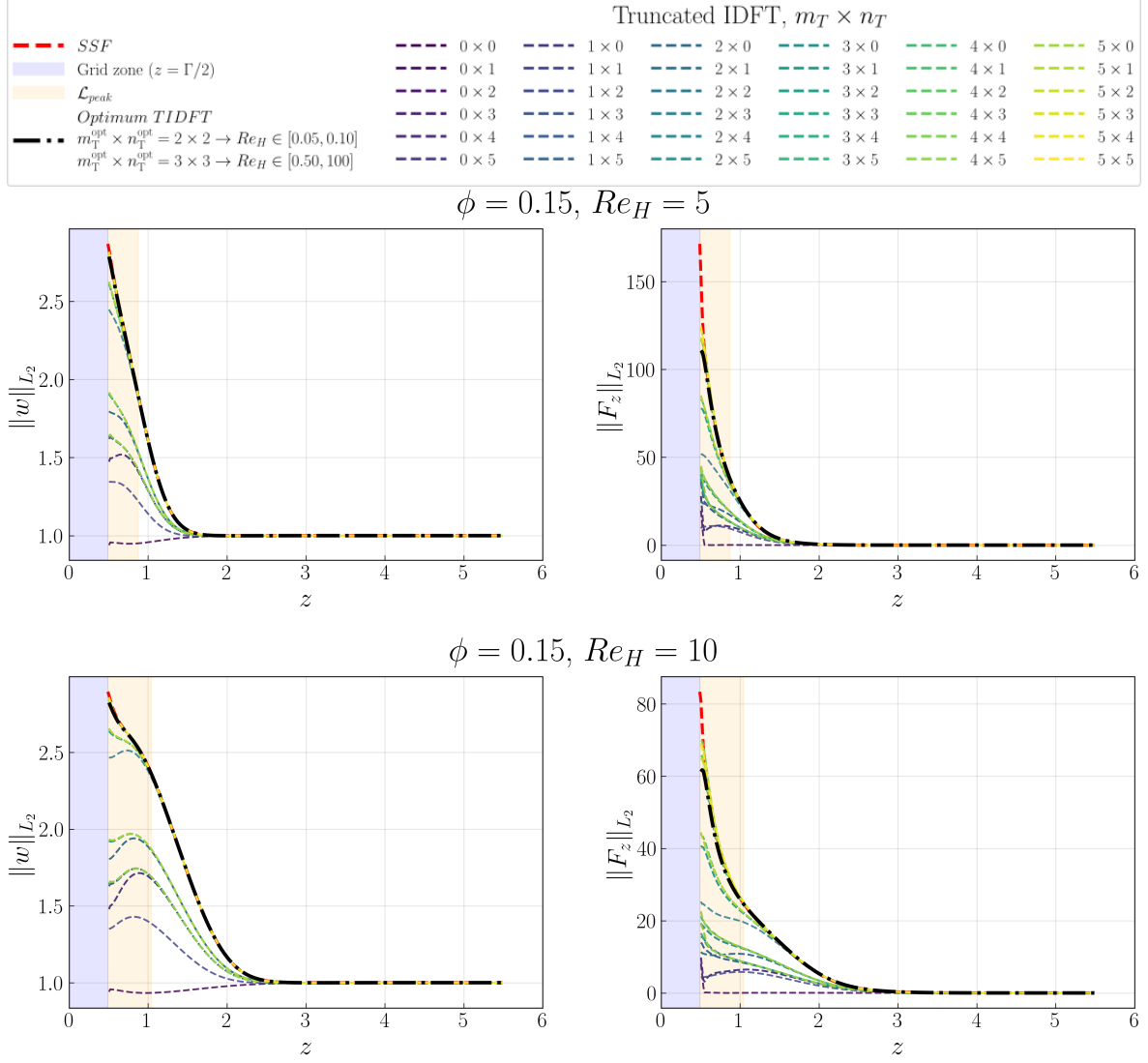


Figure B.78: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.15$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 5$ , and the lower row to  $Re_H = 10$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

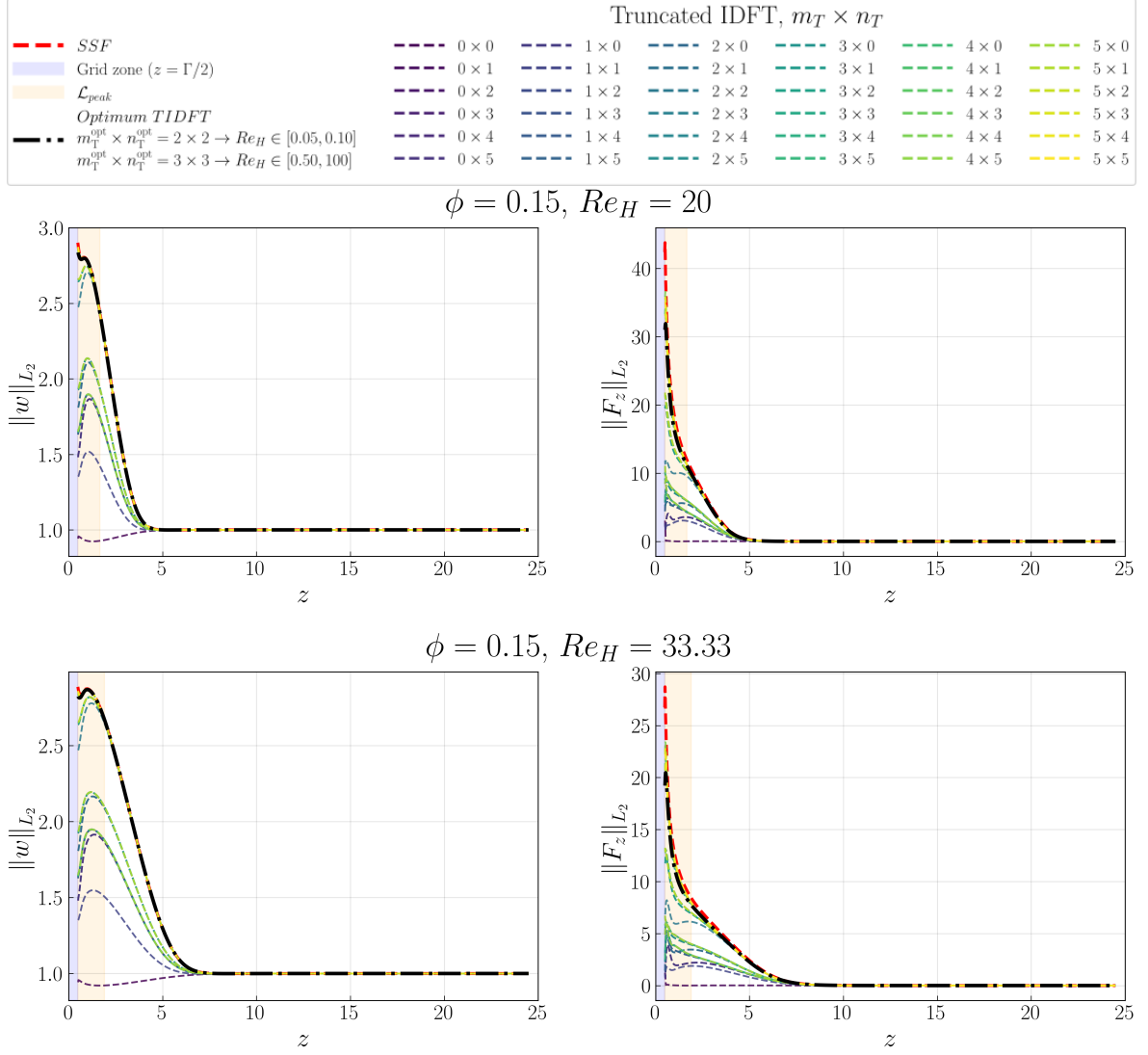


Figure B.79: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.15$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 20$ , and the lower row to  $Re_H = 33.33$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

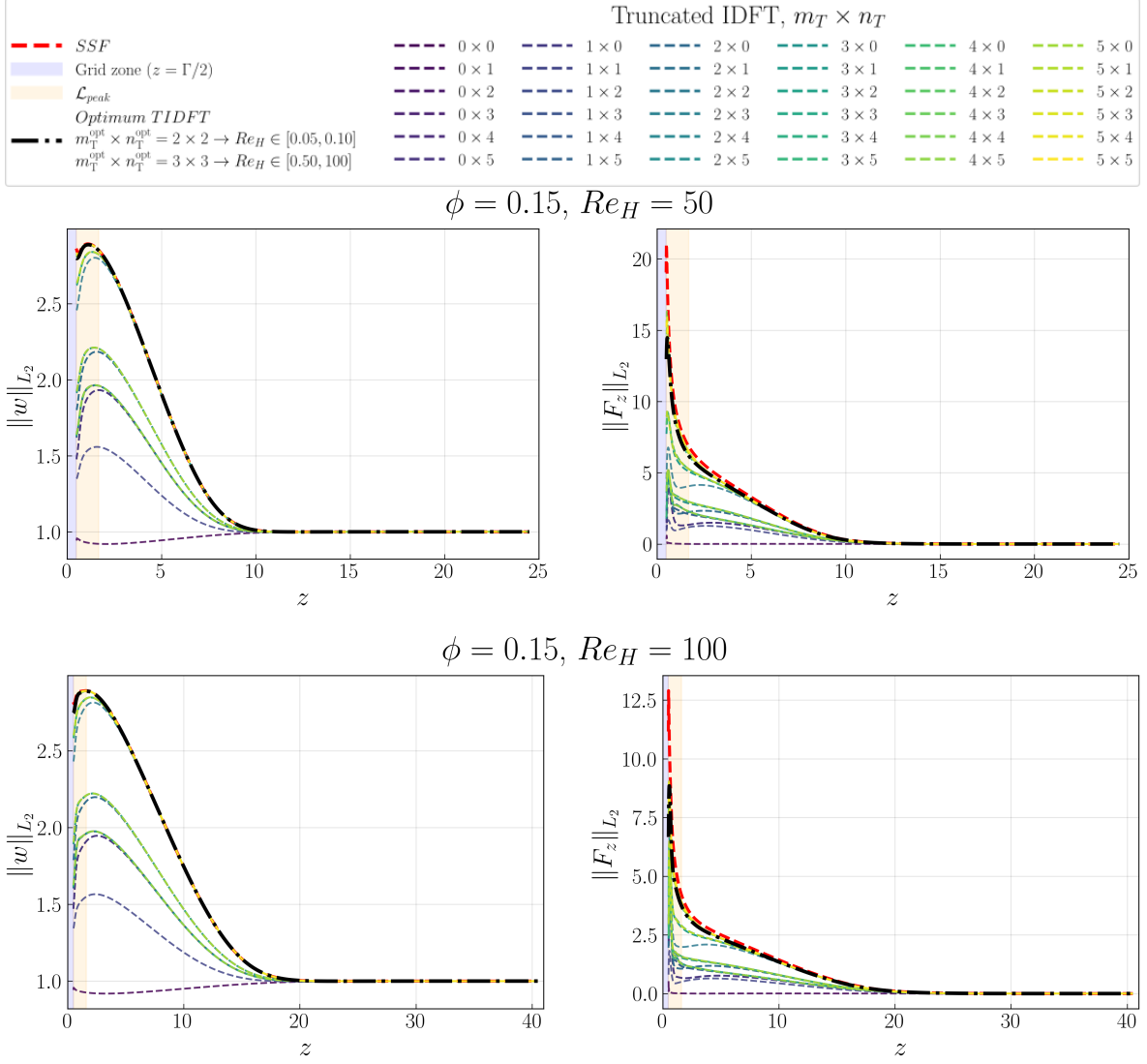


Figure B.80: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.15$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 50$ , and the lower row to  $Re_H = 100$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

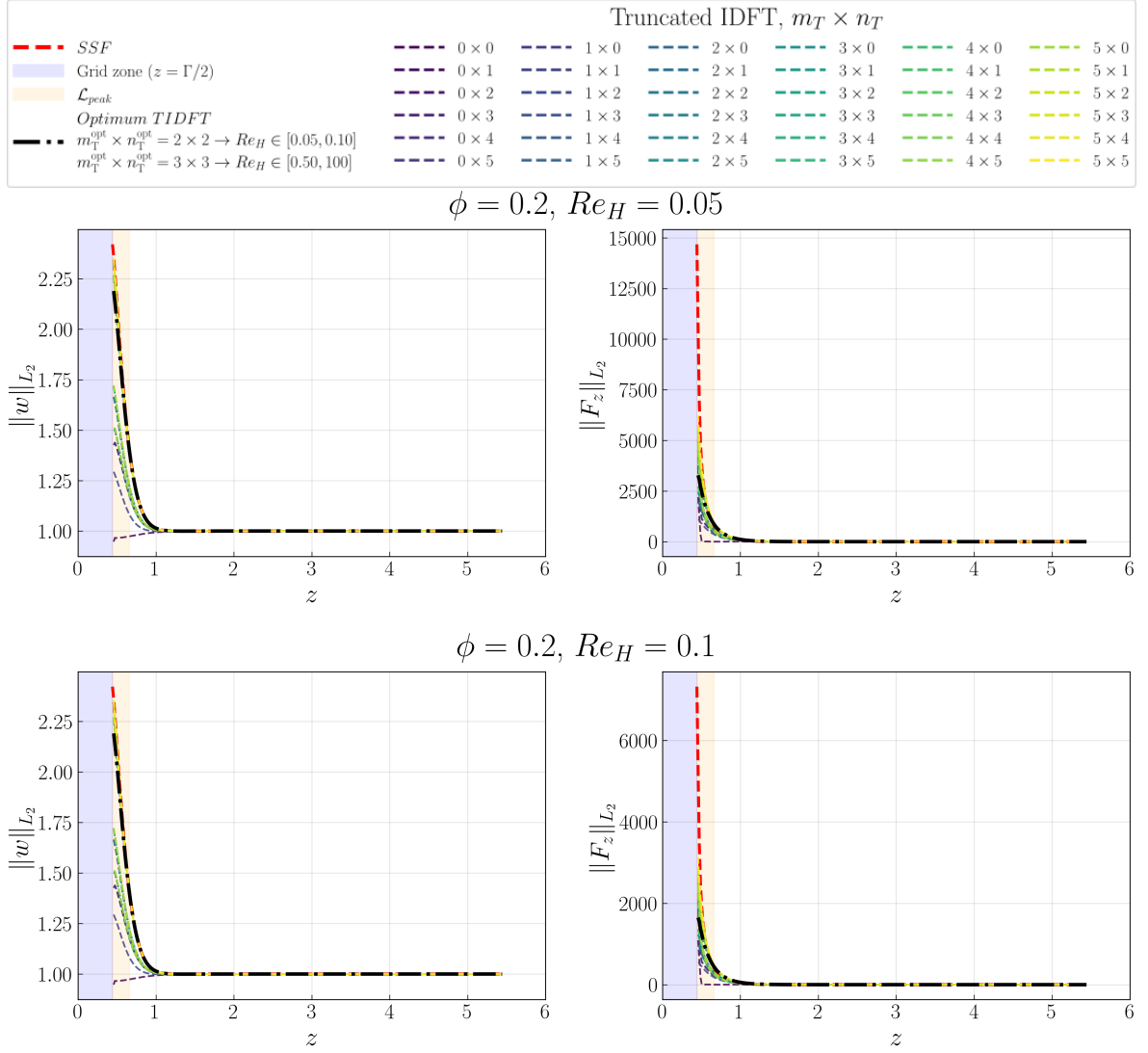


Figure B.81: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.05$ , and the lower row to  $Re_H = 0.1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

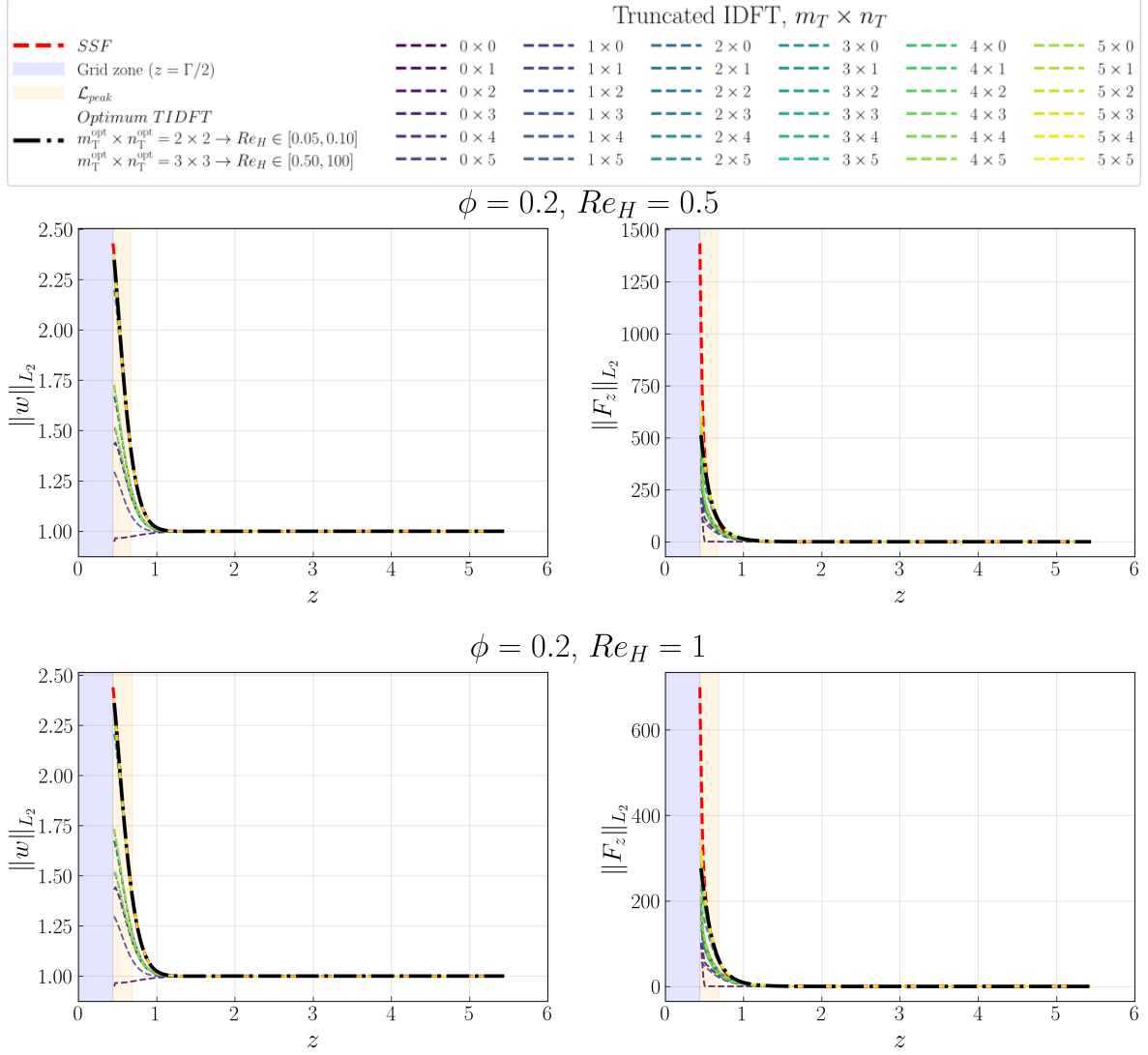


Figure B.82: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.5$ , and the lower row to  $Re_H = 1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

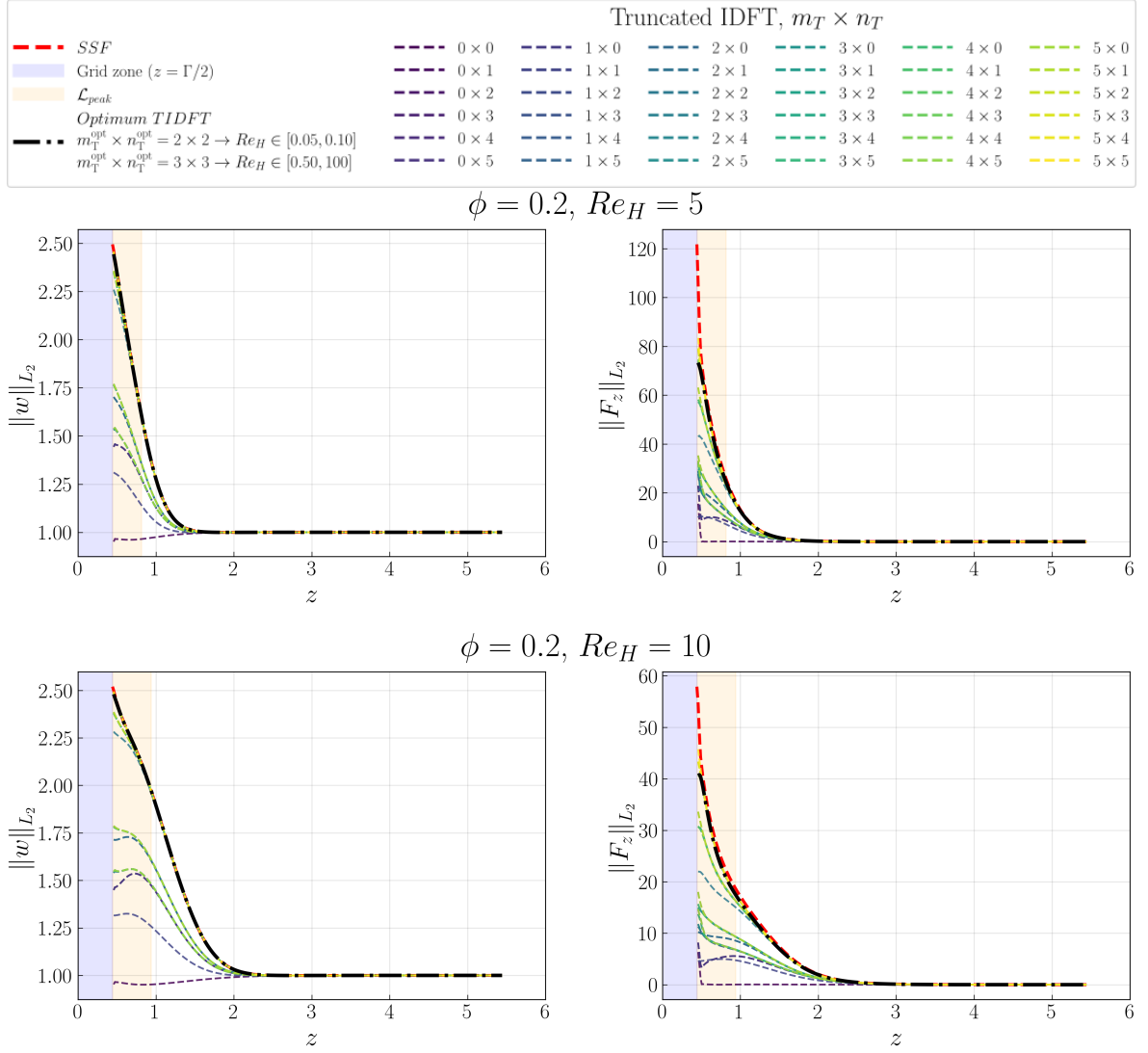


Figure B.83: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 5$ , and the lower row to  $Re_H = 10$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

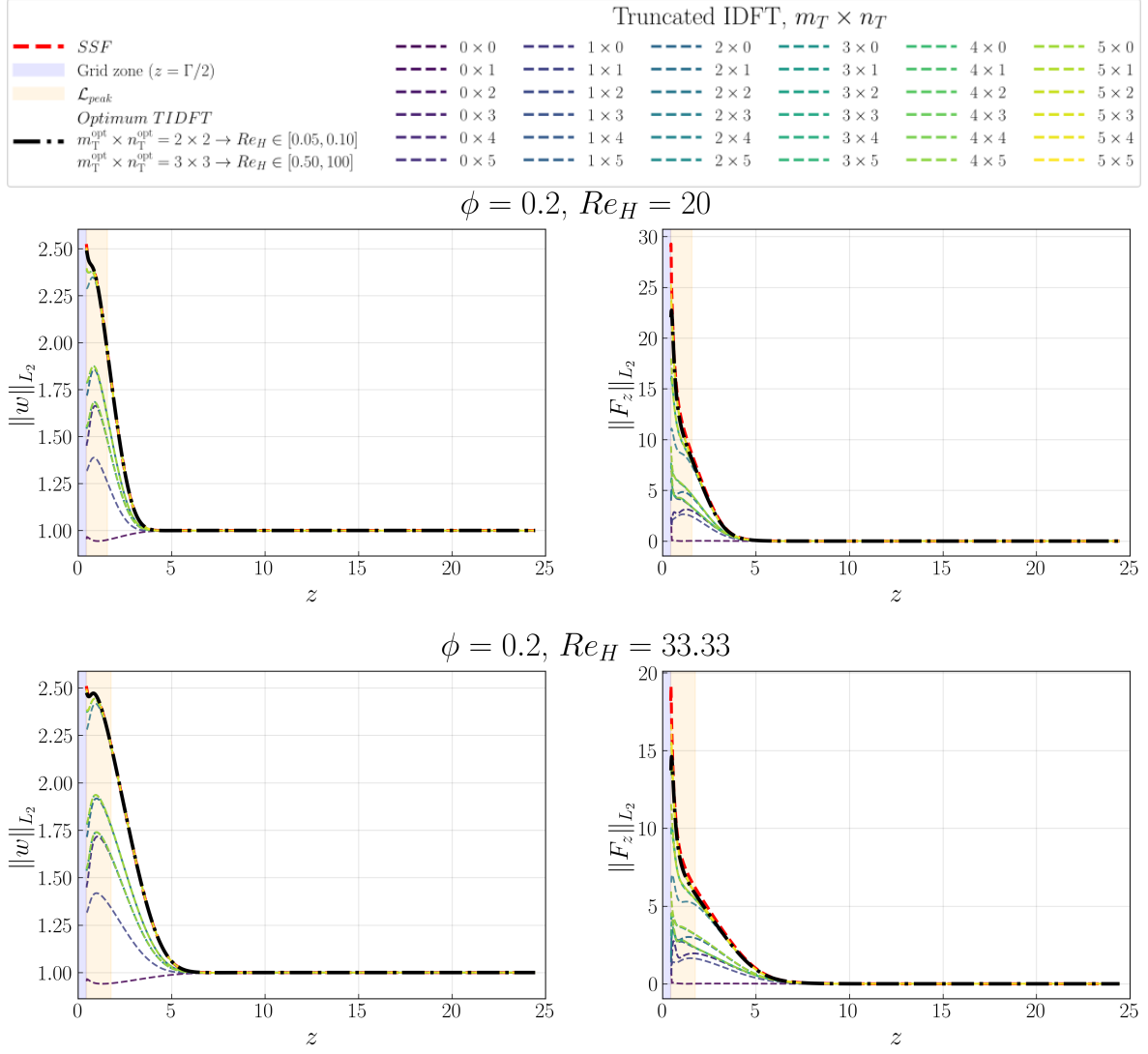


Figure B.84: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 20$ , and the lower row to  $Re_H = 33.33$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .



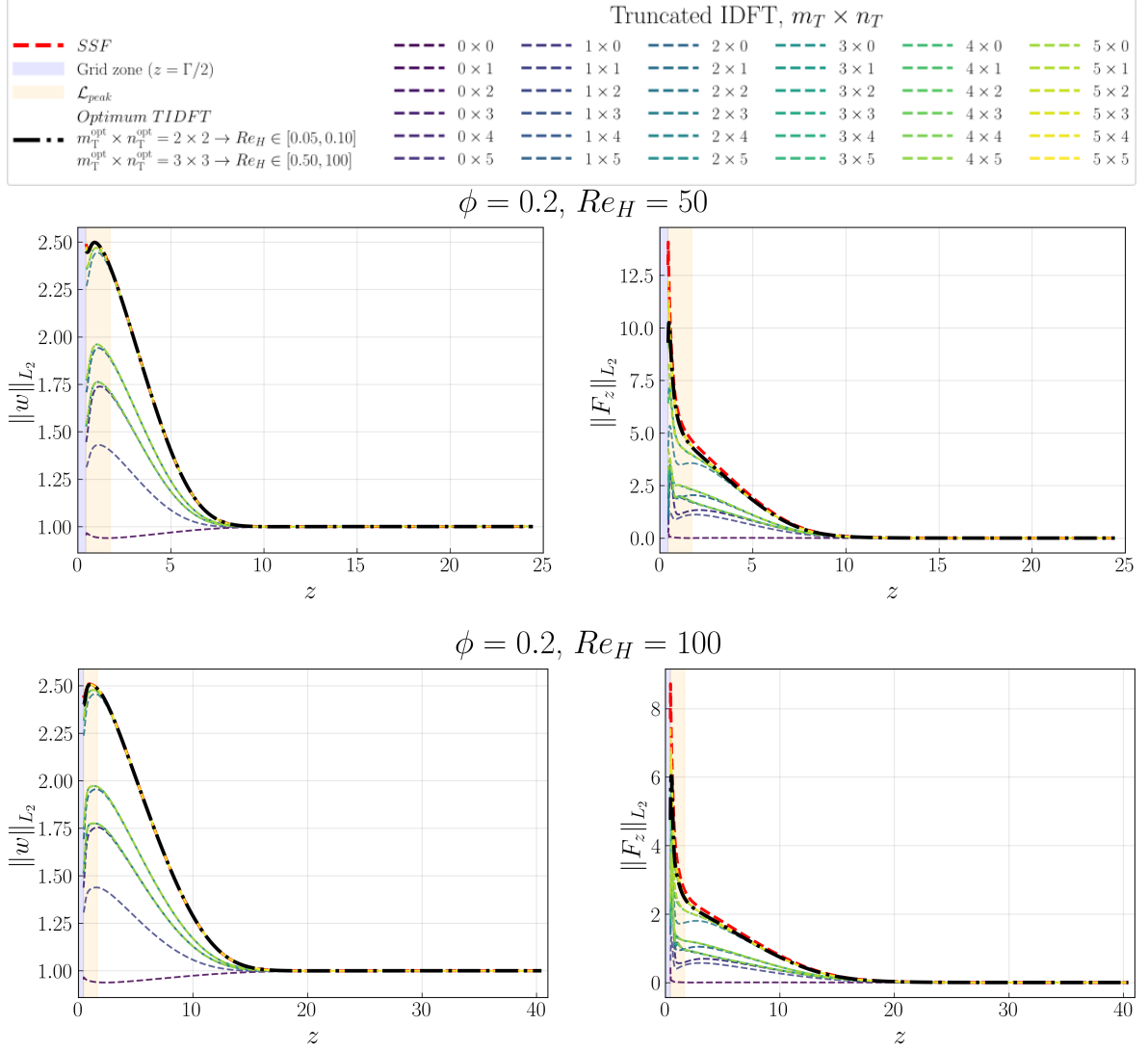


Figure B.85: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 50$ , and the lower row to  $Re_H = 100$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

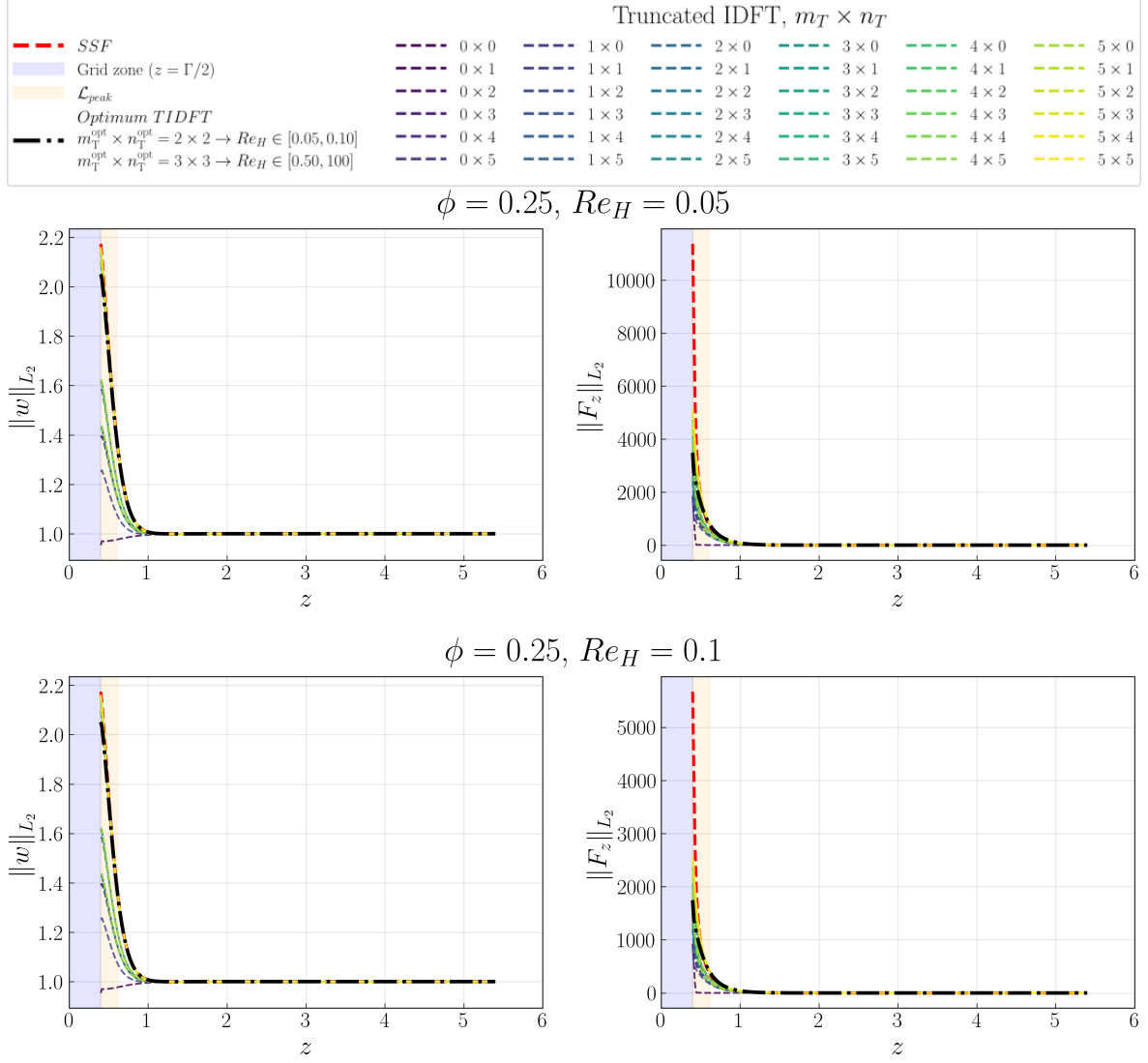


Figure B.86: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.25$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.05$ , and the lower row to  $Re_H = 0.1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

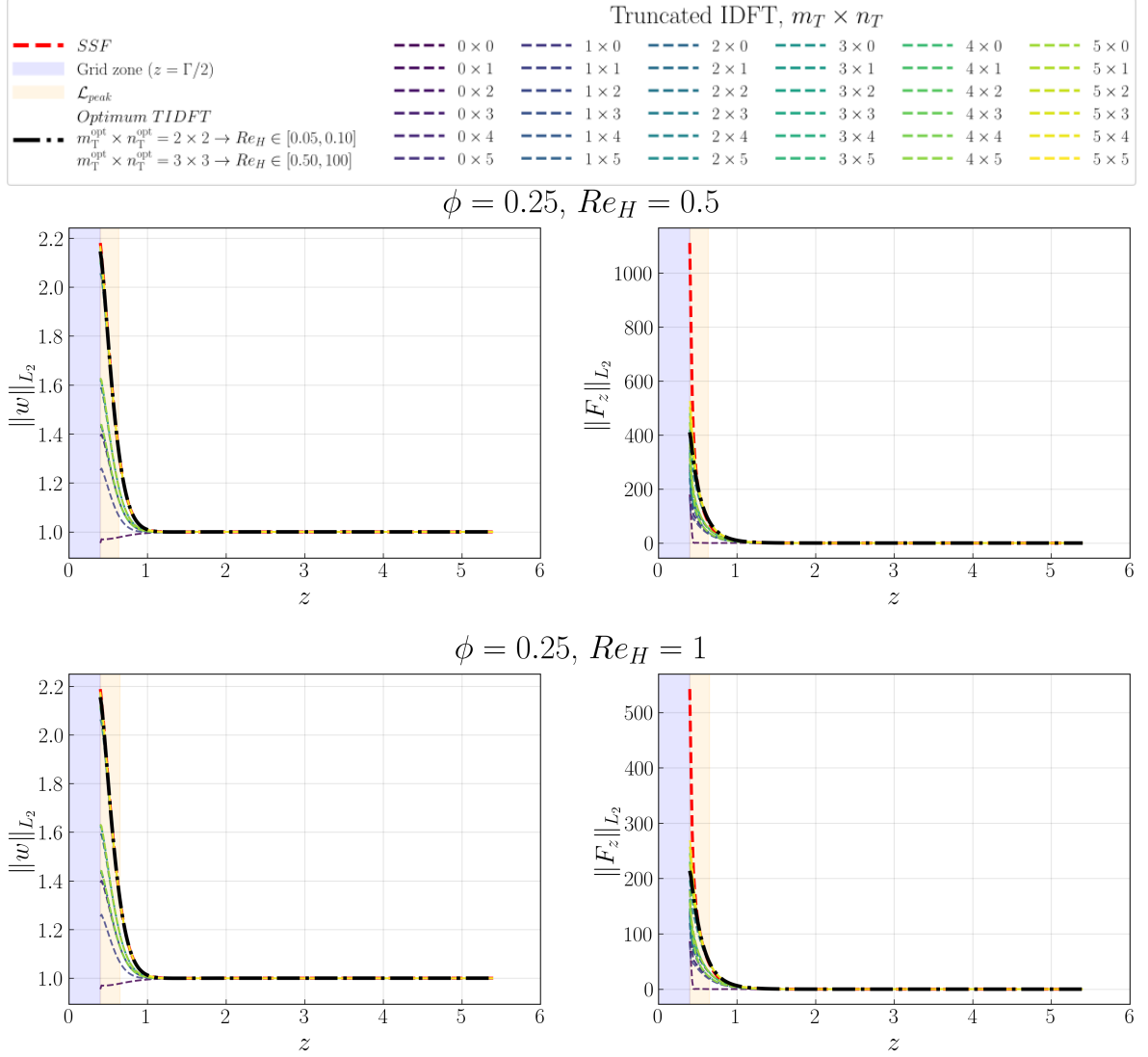


Figure B.87: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.25$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 0.5$ , and the lower row to  $Re_H = 1$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

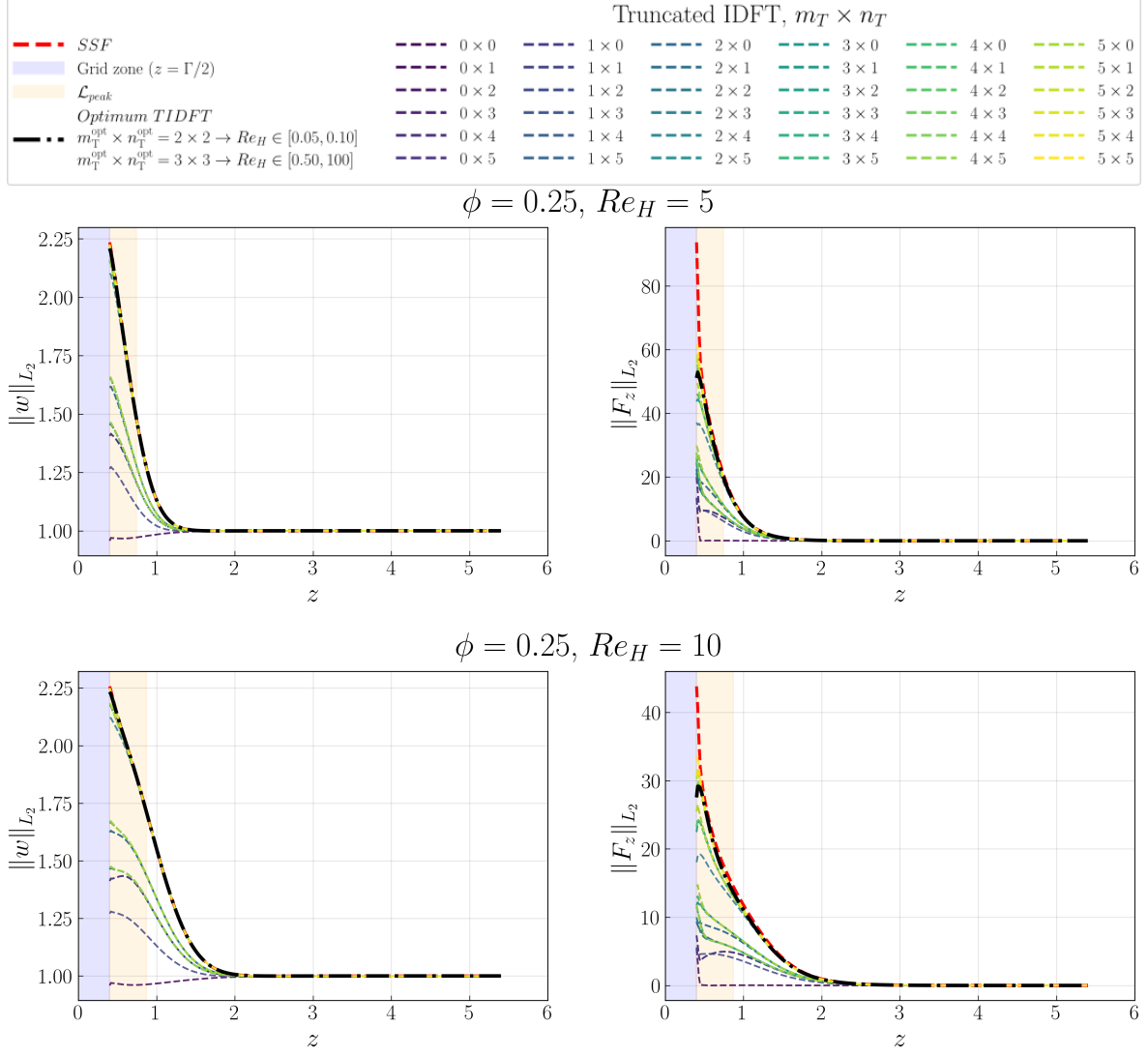


Figure B.88: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.25$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 5$ , and the lower row to  $Re_H = 10$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

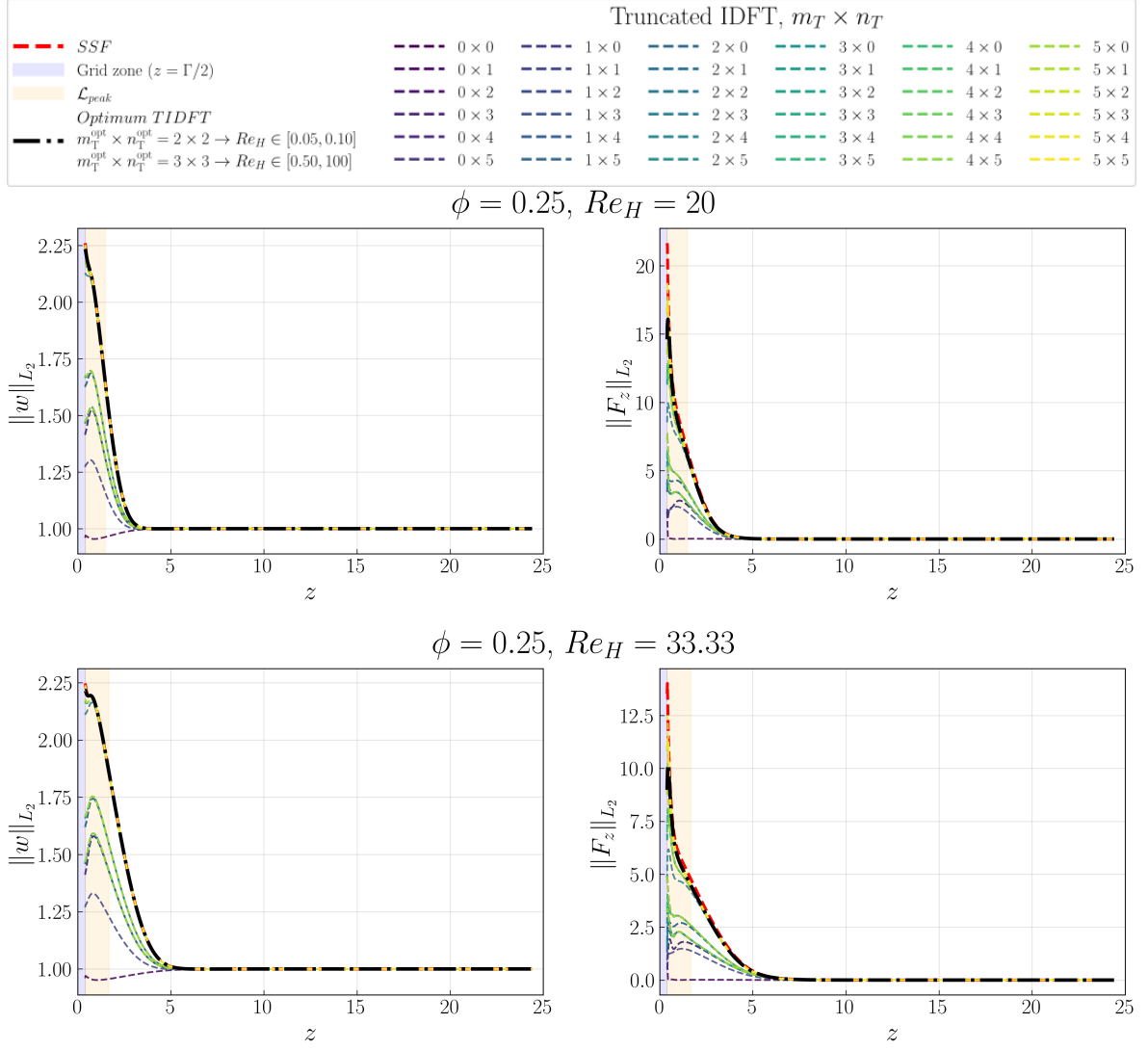


Figure B.89: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.25$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 20$ , and the lower row to  $Re_H = 33.33$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

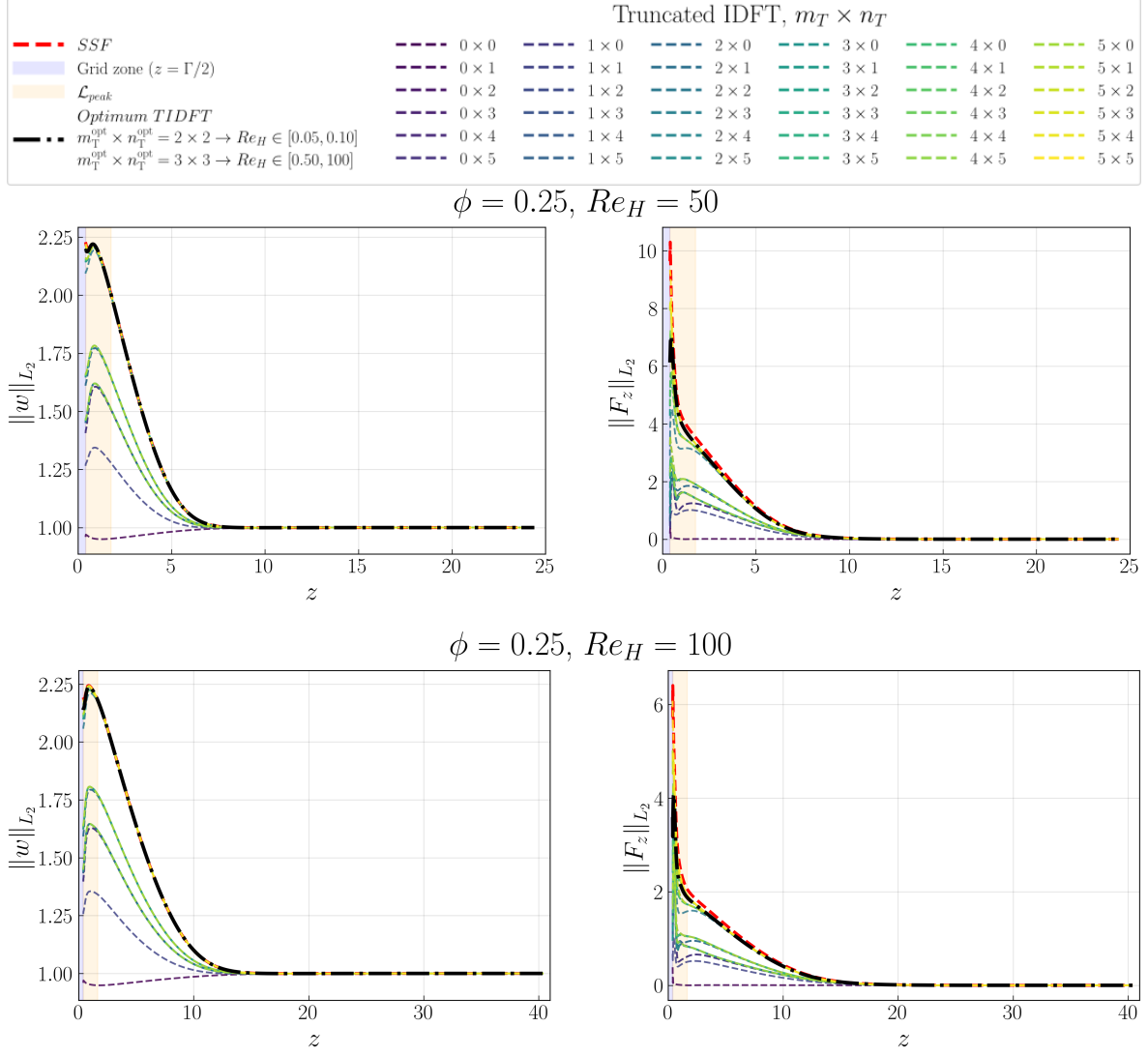


Figure B.90: Streamwise evolution of the cross-sectional  $L_2$ -norms of the streamwise velocity  $w$  (left panels) and forcing  $F_z$  (right panels) for a  $1 \times 1$  SRG at  $\phi = 0.25$  and  $\alpha = 0^\circ$ . The upper row corresponds to  $Re_H = 50$ , and the lower row to  $Re_H = 100$ . Red dashed curves show the SSF reference; colored dashed curves show TIDFT reconstructions for the modal truncations listed in the shared legend. Black dash-dotted curves indicate the selected optimal truncation. The blue shaded region marks the grid, and the yellow dashed line marks  $z = \Gamma/2 + \mathcal{L}_{peak}$ .

---

Low modal truncations underestimate the reconstructed streamwise velocity and forcing near the grid, with a larger discrepancy for  $F_z$ . As more modes are retained, both fields approach the geometry-resolved SSF reference. The selected truncation closely follows the SSF curves in the effective downstream region  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . This indicates that relatively few modes capture the downstream redistribution, whereas the near-grid forcing requires greater modal resolution.

To quantify reconstruction accuracy across the investigated cases, the normalized relative error for each scalar field  $q \in \{u, v, w, F_z\}$  is defined as

$$\epsilon_q^{\text{rel}}(\phi, \text{Re}_H; m_T, n_T) = \frac{\|q_{\text{SSF}} - q_{\text{TIDFT}}\|_{L_2(z_{\text{eff}})}}{\|q_{\text{SSF}}\|_{L_2(z_{\text{eff}})}}, \quad z_{\text{eff}} = [\Gamma/2 + \mathcal{L}_{\text{peak}}, z_{\text{max}}]. \quad (\text{B.10})$$

Here,  $z_{\text{max}}$  is the numerical-domain outlet. Evaluating the error over  $z_{\text{eff}}$  excludes the localized near-grid build-up and focuses on the downstream redistribution.

The following figures report  $\epsilon_q^{\text{rel}}$  as a function of  $(m_T, n_T)$  for each indicated porosity and Reynolds number.

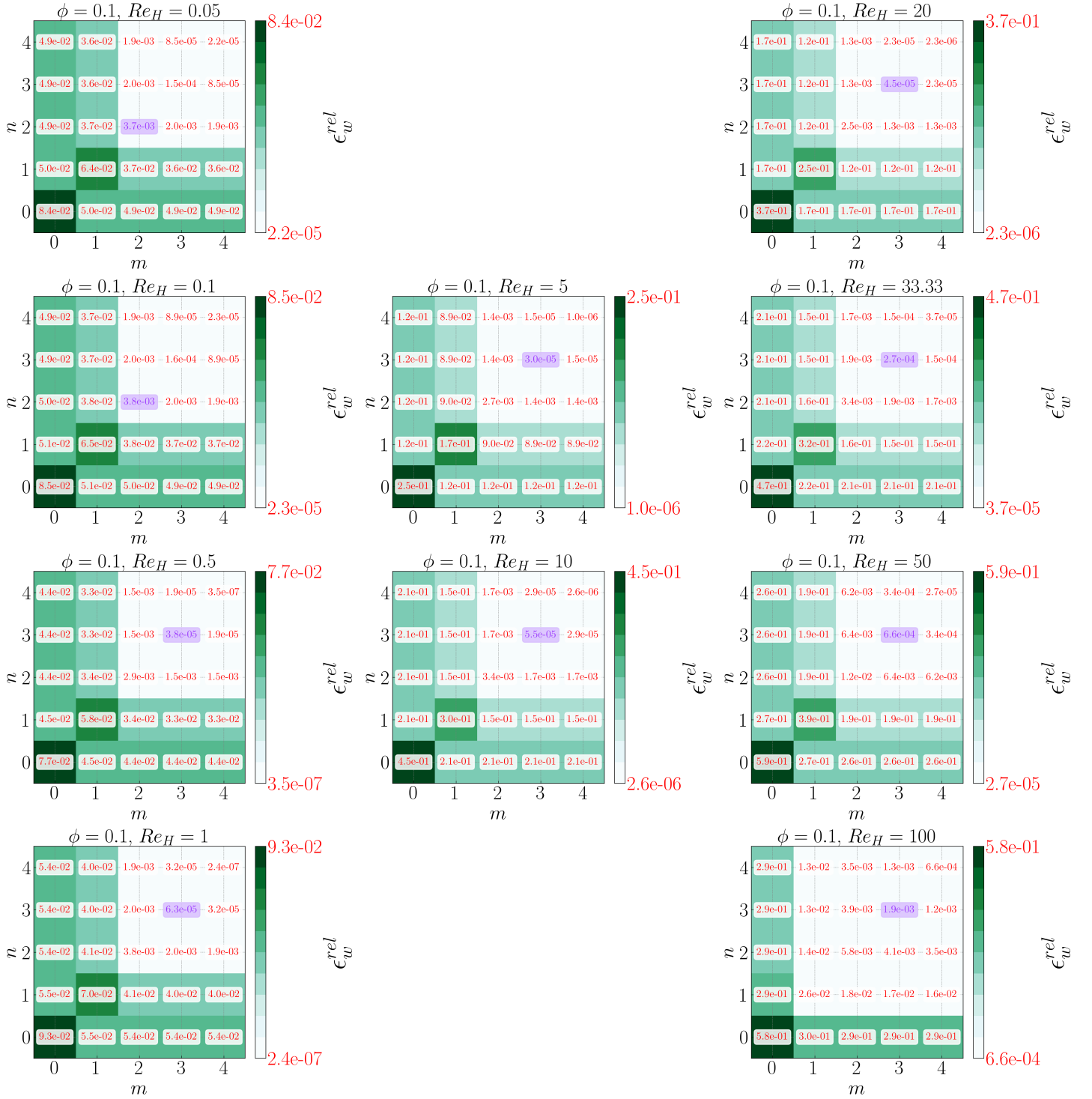


Figure B.91: Relative reconstruction error of the streamwise velocity component  $w$  obtained using Truncated Inverse Discrete Fourier Transform (TIDFT) representations for the  $1 \times 1$  SRG at  $\phi = 0.1$ . Each panel shows  $\epsilon_w^{\text{rel}}$  as a function of the retained modal truncation ( $m_T, n_T$ ) at the indicated Reynolds number. The error is evaluated within the effective downstream redistribution region,  $z \geq \Gamma/2 + \mathcal{L}_{\text{peak}}$ . Together, the panels assess the modal support required to reconstruct the corresponding downstream field over the investigated Reynolds-number range.



